



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

ENEE4113 - COMMUNICATION LAB

Report#3

Experiment 10: Amplitude Shift Keying (ASK)

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Abstract

The experiment focuses on Amplitude Shift Keying (ASK) modulation in both the time and frequency domains. It explores two keying methods, Hard and Soft keying, and examines the effect of the modulation index. The experiment involves detecting the comparator threshold, demodulation using both hard and soft keying techniques, and studying the impact of changing the message signal's amplitude, frequency, and duty cycle. By analyzing these parameters, students gain practical understanding of ASK modulation and its characteristics.

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1.Theory

1.1 Amplitude shift keying (ASK) Modulation:

A high frequency carrier is present in any modulated transmission. When ASK modulated, the binary signal produces a zero value for low input and the carrier output for high input.

The ASK modulated waveform and its input are depicted in the image below.

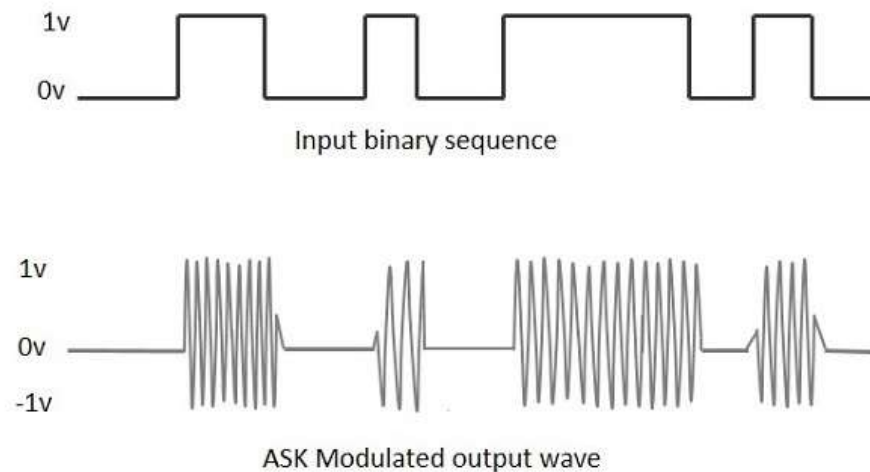


Figure 1: The ASK modulated waveform and its input

To understand how to produce this ASK modulated wave, we must first understand how the ASK modulator works.

❖ ASK Modulator

The carrier signal generator, the binary sequence from the message signal, and the band-limited filter are shown in the ASK modulator block diagram. The ASK Modulator's block diagram is shown below:

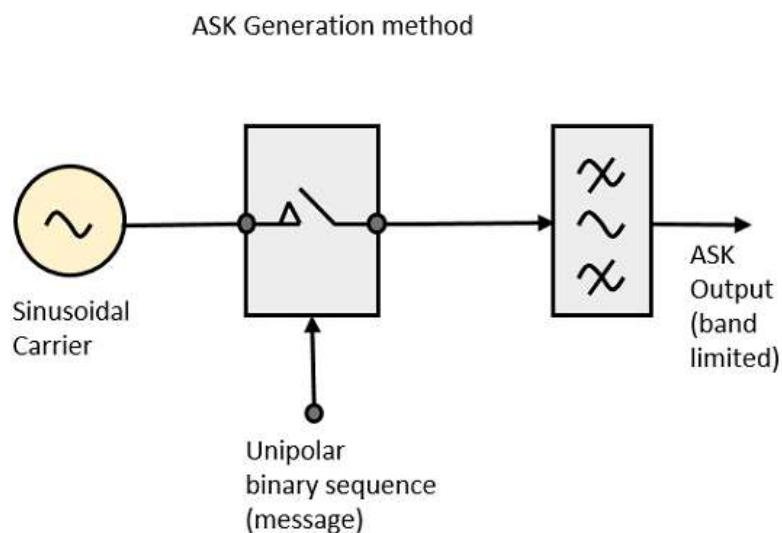


Figure 2: The ASK Modulator's block diagram

A continuous high-frequency carrier is sent by the carrier generator. The message signal's binary sequence causes the unipolar input to be either High or Low. The high signal shuts the switch, enabling a carrier wave to pass through. As a result, for high input, the output will be the carrier signal. When the input is low, the switch opens, preventing any voltage from appearing. As a result, the production will be modest.

The band-limiting filter or the pulse-shaping filter molds the pulse based on the amplitude and phase parameters of the band-limiting filter or the pulse-shaping filter.

Amplitude-shift keying (ASK) is a kind of amplitude modulation in which digital data is represented as fluctuations in the amplitude of a carrier wave. A symbol representing one or more bits is delivered in an ASK system by delivering a fixed-amplitude carrier wave at a given frequency for a specific time length. For example, if each symbol represents a single bit, the carrier signal might be carried at full amplitude when the input value is 1, but at decreased amplitude or not at all when it is 0.

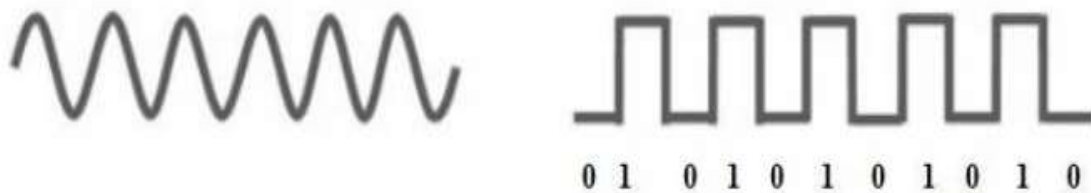
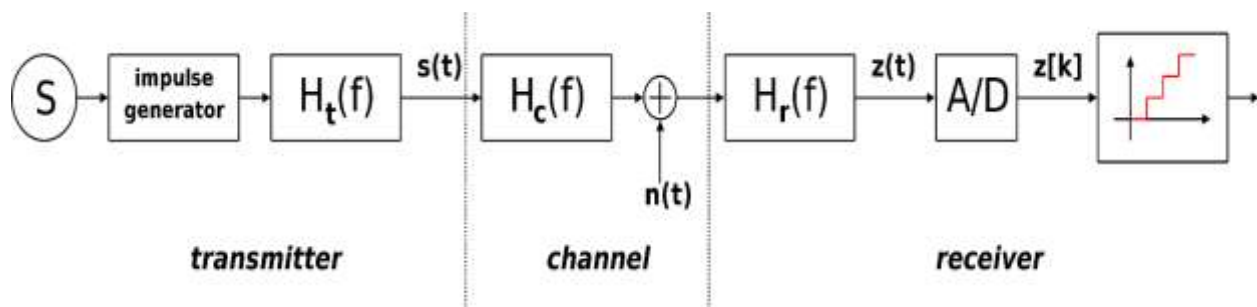


Figure 3: analog-and-digital-signals

To represent digital data, any digital modulation system employs a finite number of different signals. ASK employs a limited number of amplitudes, each with its own unique pattern of binary bits. Typically, each amplitude encodes the same number of bits. Each bit pattern generates the symbol represented by the given amplitude. The demodulator, which is tailored to the modulator's symbol set, determines the amplitude of the received signal and maps it back to the symbol it represents, therefore retrieving the original data. The carrier's frequency and phase are kept constant.

More advanced encoding systems that encode data in groups utilizing extra amplitude levels have been developed. A four-level encoding method, for example, may represent two bits with each amplitude shift; an eight-level scheme can represent three bits; and so on. Because much of the data is delivered at low power, many types of amplitude-shift keying require a high signal-to-noise ratio for recovery.



1.2 Amplitude shift keying (ASK) demodulation:

ASK Demodulation methods are classified into two categories. They have

- Demodulation/detection of asynchronous ASK
- Demodulation/detection of Synchronous ASK

When the clock frequency at the transmitter matches the clock frequency at the receiver, the approach is known as synchronous, because the frequency is synchronized. Otherwise, it is referred to as asynchronous.

❖ Asynchronous ASK Demodulator

The Asynchronous ASK detector consists of a half-wave rectifier, a low pass filter, and a comparator. Following is the block diagram for the same.

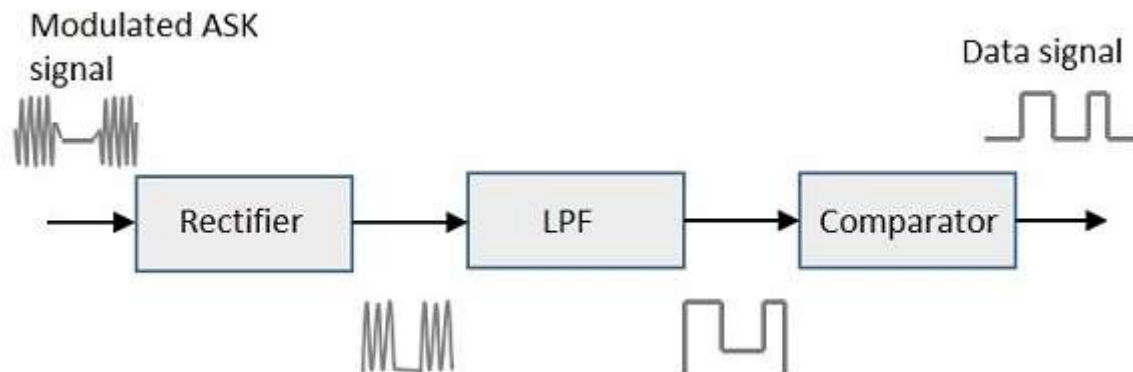


Figure 4: Asynchronous ASK detector

The modulated ASK signal is sent into a half-wave rectifier, which produces a positive half-wave output. The low pass filter suppresses the higher frequencies and provides an envelope sensed output to the comparator, which produces a digital output.

❖ Synchronous ASK Demodulator

Synchronous ASK detector consists of a Square law detector, low pass filter, a comparator, and a voltage limiter. Following is the block diagram for the same.

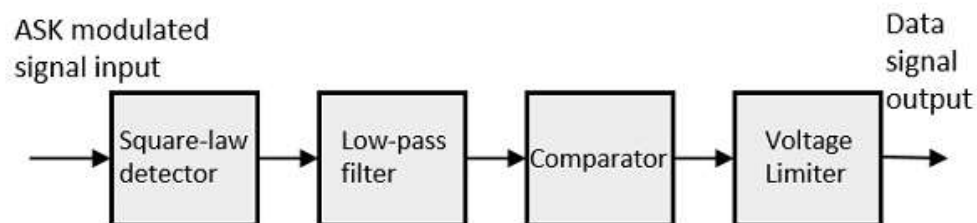


Figure 5: Synchronous ASK detector

The Square law detector receives the ASK modulated input signal. The output voltage of a square law detector is proportional to the square of the amplitude modulated input voltage. Higher frequencies are reduced by the low pass filter. The comparator and voltage limiter aid in obtaining a clean digital output.

1.3 The difference between Hard Keying & Soft Keying:

Amplitude Shift Keying (ASK) modulation employs two separate techniques: hard keying and soft keying. The binary information is directly imprinted onto the carrier signal in hard keying by switching between two amplitude levels. This technique is based on discrete amplitude changes that represent one and zero. Soft keying, on the other hand, uses continuous modulation of the amplitude of the carrier signal to transfer binary information. This produces a smooth transition between two amplitudes, allowing for greater flexibility in data representation. The decision between hard and soft keying is influenced by factors such as bandwidth needs, signal quality, and noise sensitivity.

2. Procedure & Results

Part 1: Amplitude shift keying Modulation (Time domain) Hard Keying

The components were assembled as shown below.

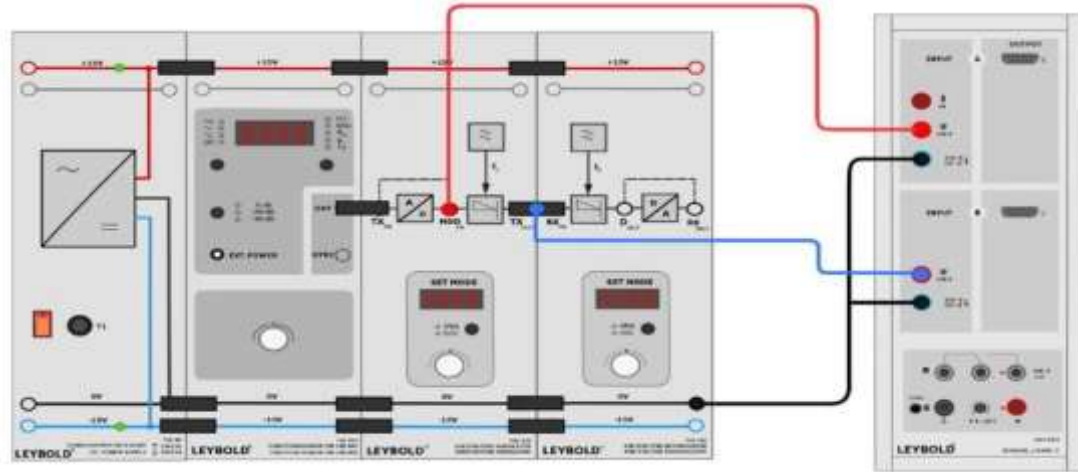


Figure 6: Assembling Components for ASK Modulation

The function generator was set to create a square wave with a frequency of 1000 Hz, a V_{ss} of 10V (amplitude of 5V), and a duty cycle of 50% in this section. The modulator mode was set to Digital (DIG), and the modulation technique Amplitude Shift Keying - Hard keying was used. The modulation index, m , was set to one hundred percent. The Cassy sensor UA1 was attached to the socket MODIN to capture the unipolar square-wave signal, while the Cassy sensor UB1 was linked to the modulator's output TXOUT to capture the ASK signal.

❖ Then the measurement had begun and the output appeared as follow:

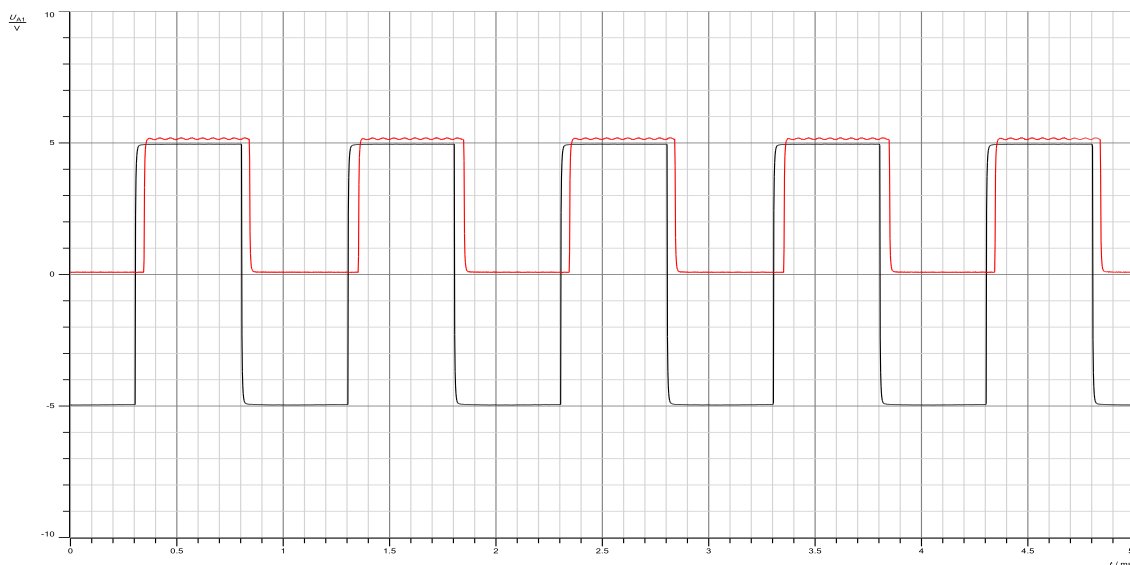


Figure 7: Message Signal in Time Domain

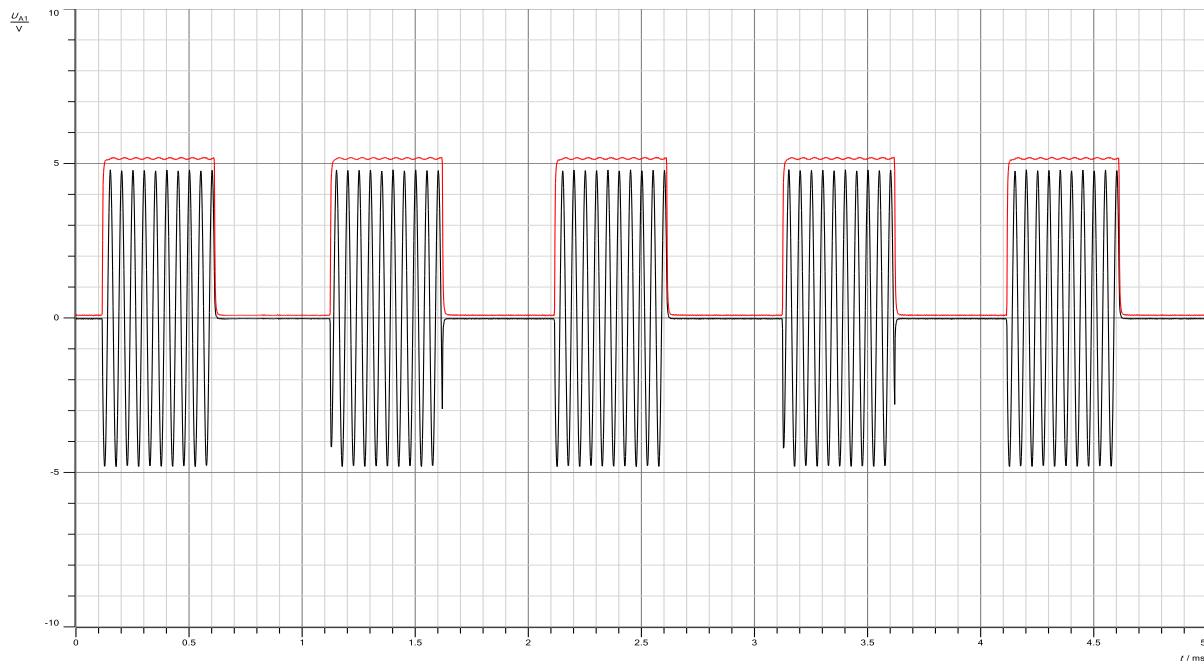


Figure 8: ASK modulated signal in time domain $m= 100\%$ hard keying

The previous part was still the same, but with a different amplitude shift keying - Hard keying was used with a modulation index m of 50%. The measurements had begun. The figure below depicts the ASK modulated signal in relation to the message signal.

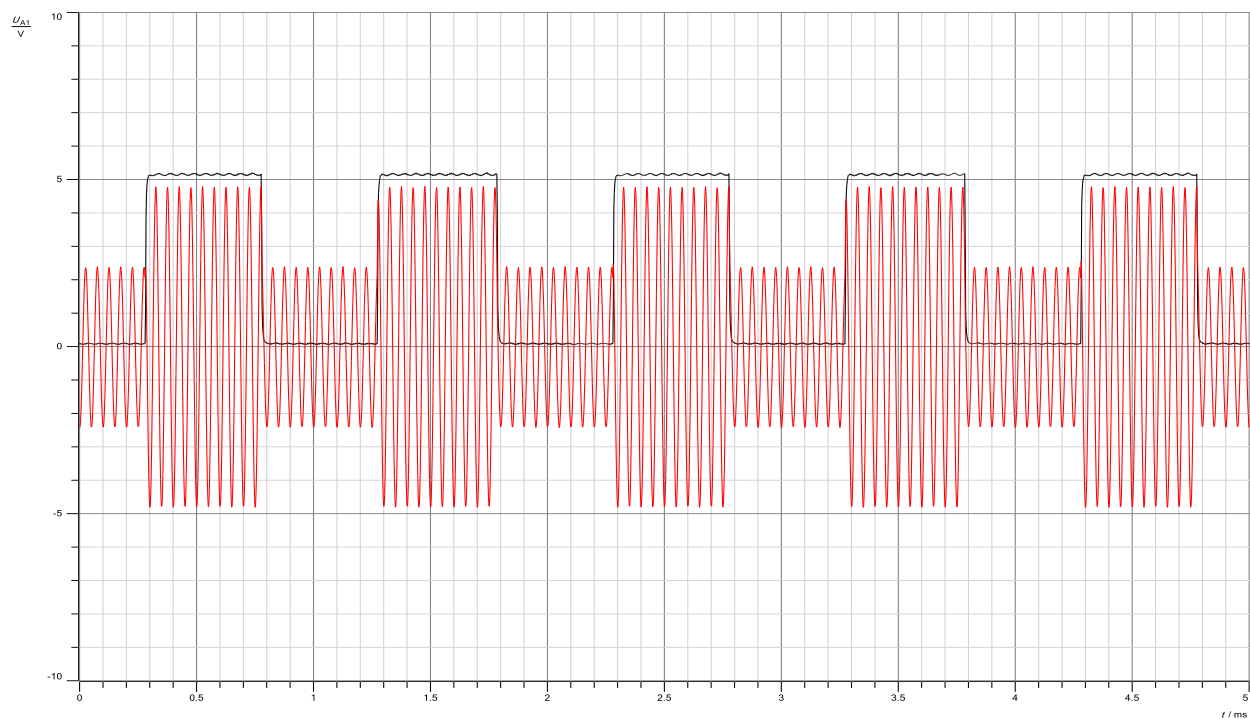


Figure 9: ASK modulated signal in time domain $m = 50\%$ hard keying

By observing the provided figures, it becomes evident that a modulation index of 50% ($m = 0.5$) produces a moderate change in the amplitude of the modulated signal. However, with a modulation index of 100% ($m = 1$), the carrier signal experiences complete on-off keying. In the case of hard keying, there are no intermediary amplitude levels present in ASK-H; the signal is either fully "on" or "off."

- The modulation scheme employed in this experiment is unipolar non-return-to-zero (NRZ) modulation.

❖ Soft keying:

The function generator was configured to generate a square wave with a frequency of 1000 Hz, V_{ss} of 10V (amplitude of 5V), and a duty cycle of 50%. The modulator mode was set to Digital (DIG). For modulation, Amplitude Shift Keying – Soft keying technique was chosen, with a modulation index of 100% (A_12). To capture the signals, the Cassy sensor UA1 was connected to the socket MODIN, capturing the unipolar square-wave message signal, while the Cassy sensor UB1 was connected to the output TXOUT of the modulator, capturing the ASK modulated signal.

The measurement process was initiated, and was showed as follow:

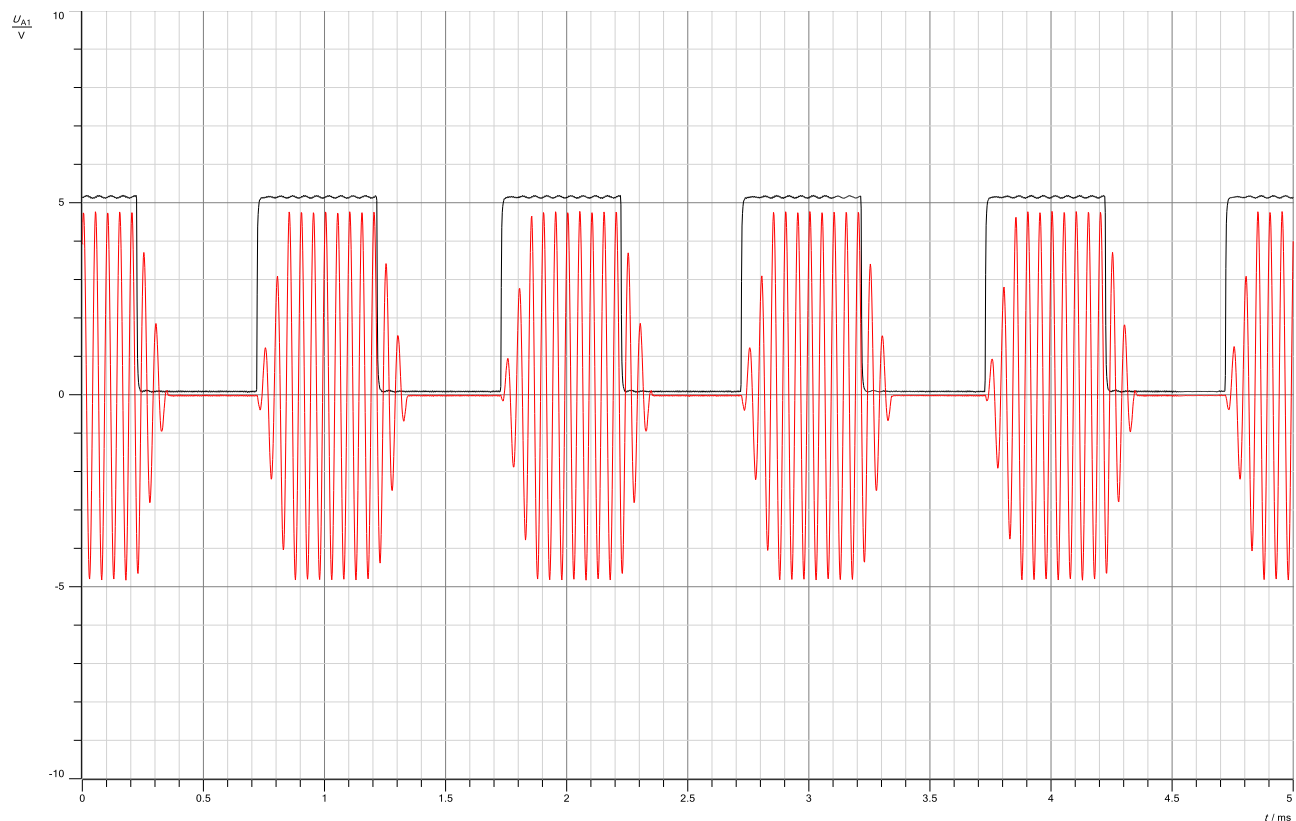


Figure 10: ASK modulated signal in time domain $m = 100\%$ soft keying

The previous part was still the same, but with a different amplitude shift keying – Soft keying was used with a modulation index m of 50%. The measurements had begun. The figure below depicts the ASK modulated signal in relation to the message signal.

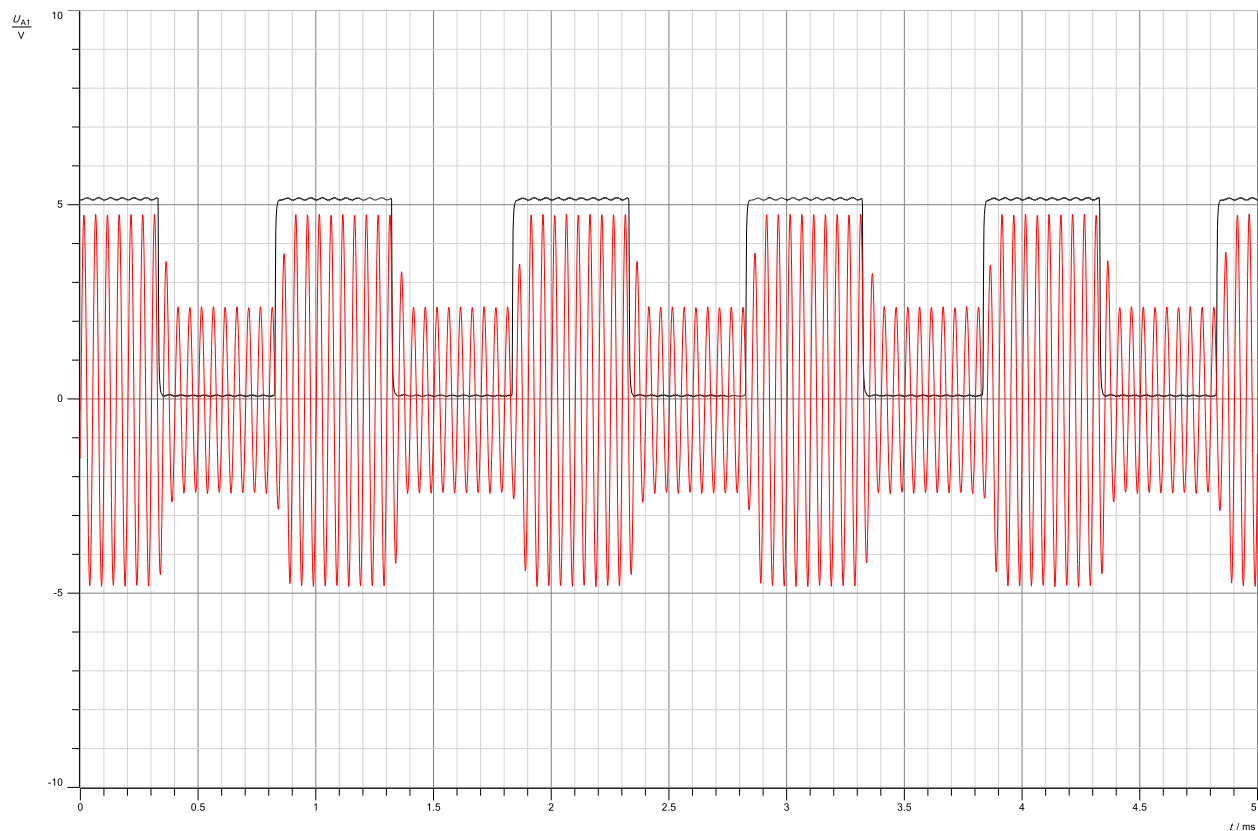


Figure 11: ASK modulated signal in time domain $m = 50\%$ soft keying

Comparing the figures, A modulation index of 50% ($m = 0.5$) causes a moderate amplitude variation in the modulated signal, while a modulation index of 100% ($m = 1$) leads to complete on-off keying of the carrier signal. Soft keying uses multiple amplitude levels to encode and transmit multiple signals per modulation symbol, unlike hard keying. Both Amplitude Modulation (AM) and Amplitude Shift Keying (ASK) change the carrier signal's strength to convey information. AM smoothly varies the carrier's strength, while ASK switches between levels. Demodulation extracts the original information from the modulated signal. Soft keying techniques have a narrower bandwidth due to smoother signal transitions, while hard keying requires a wider bandwidth due to high-frequency harmonics caused by quick transitions.

❖ Detecting the Comparator Threshold

Utilizing the identical setup and connections employed in previous sections, this portion of the experiment involved configuring the function generator to produce a DC signal with a voltage of 0V. Amplitude Shift Keying (ASK) modulation with hard keying and a modulation index of 100% (A_11) was chosen. The measurement process was initiated, and the DC value was progressively increased until the high level of the ASK signal became discernible. The recorded DC value at this particular point was identified as the upper threshold.

The figure below shown it:

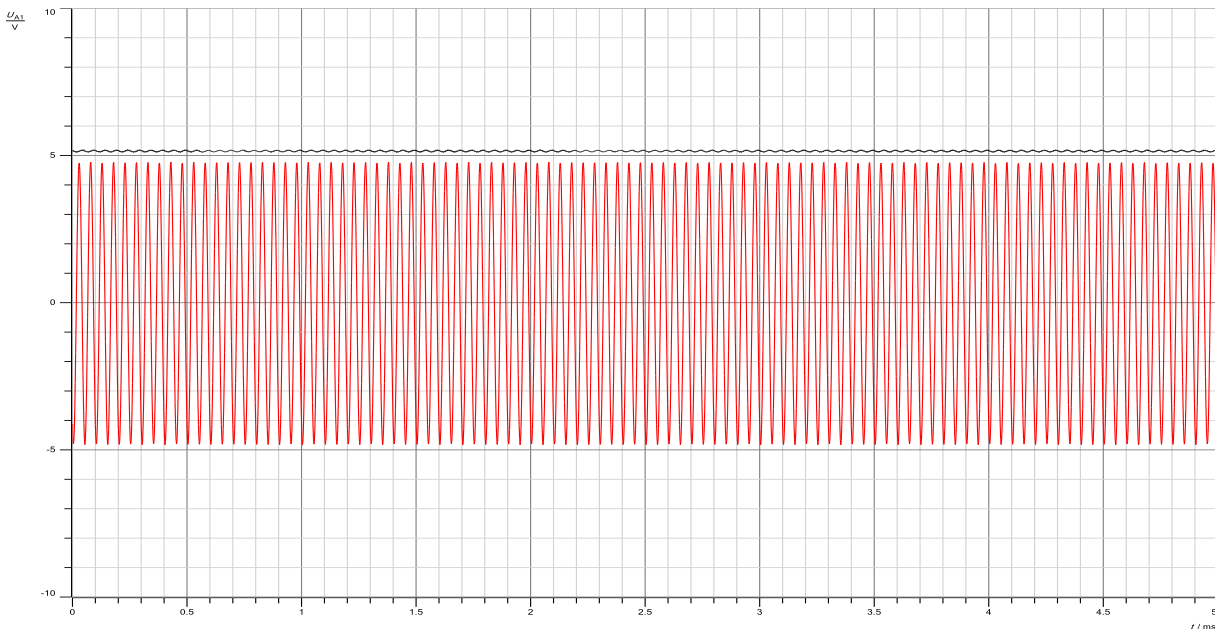


Figure 12: Detecting upper threshold

In this part of the experiment, the DC value was gradually decreased from 2.5V while closely monitoring the ASK output signal. The objective was to identify the specific DC value at which the signal transitioned to the low level (zero volts), which was recorded as the lower threshold.

The analysis revealed that the upper threshold DC value was measured at 2.15V, signifying the point at which the ASK signal switched to the high level. Similarly, the lower threshold DC value was noted as 1.68V, indicating the transition to the low level.

Within the range encompassing the high and low levels of the ASK signal, rapid changes occur as the modulating signal switches. These quick transitions can cause variations in the signal's amplitude.

Part 2: Amplitude shift keying Modulation (Frequency domain) Hard Keying

Using the same setup and connections as in previous sections, this part of the experiment involved configuring the function generator to produce a square wave with a frequency of 1KHz, an amplitude of 5V (Vss of 10V), and a duty cycle of 50%. The modulator mode was set to Digital (DIG), and Amplitude Shift Keying (ASK) with hard keying and a modulation index of 100% was selected. The Cassy sensor UA1 was connected to the unipolar square-wave signal at socket MODIN, while the Cassy sensor UB1 was connected to the ASK signal at the output TXOUT of the modulator.

The measurement had begun and the output show the ASK modulated signal in the frequency domain as follow:

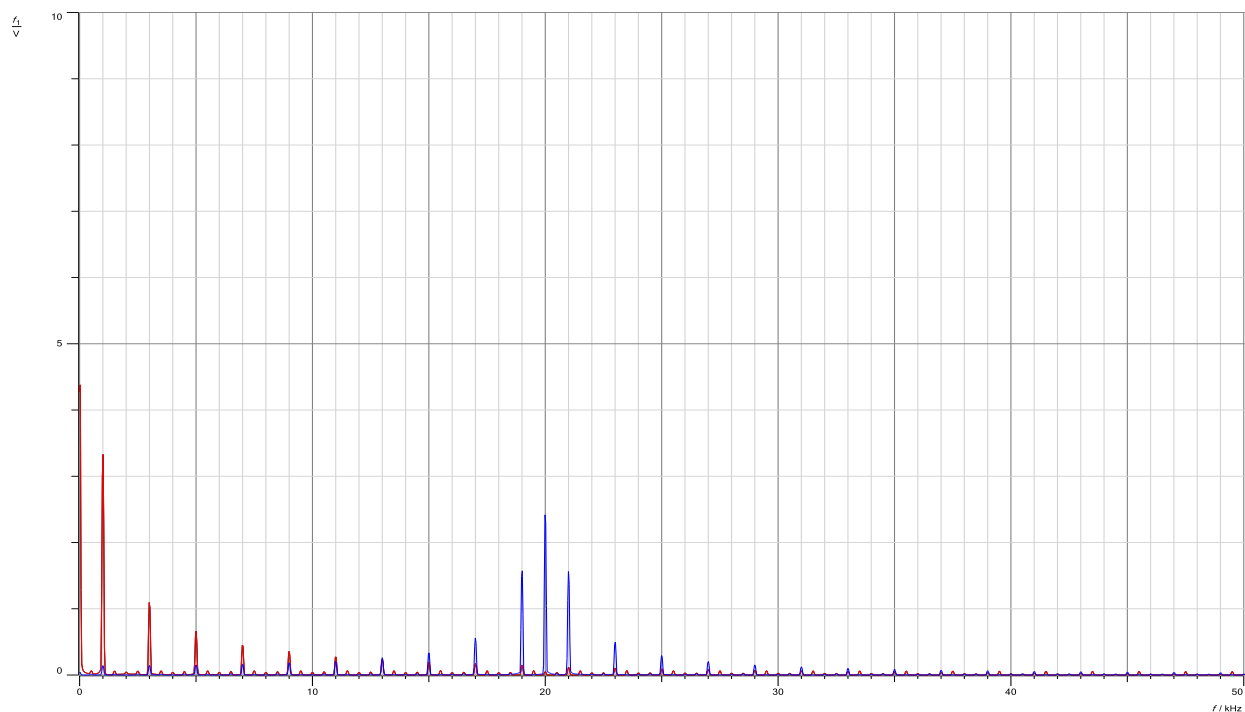


Figure 13: ASK modulated signal in frequency domain $m = 100\%$ hard keying

The figure presents the spectrum of the ASK modulated signal, revealing distinct features. The spectrum showcases a dominant carrier frequency, depicted as a single peak, along with multiple harmonics generated by the square wave message signal. These harmonics manifest as impulses separated by intervals corresponding to multiples of the carrier frequency. The amplitudes of these impulses depict the strength of the harmonic components resulting from the modulation process.

Furthermore, the red signal in the figure represents the message signal being modulated (Because the message signal is a square wave, the spectrum is a sine wave).

The previous part was still the same, but with a different amplitude shift keying – Hard keying was used with a modulation index m of 50%. The measurements had begun.

The figure below depicts the ASK modulated signal in relation to the message signal.

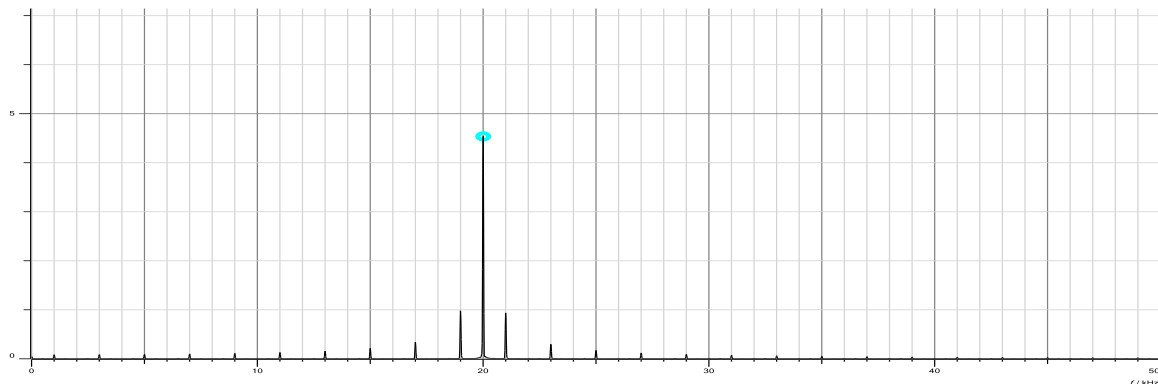


Figure 14: ASK message signal in frequency domain $m = 50\%$ hard keying

The figures show significant changes in the ASK message signal spectrum. The spectrum with a modulation index of 100% has larger impulse amplitudes and a broader bandwidth than the spectrum with a modulation value of 50%. As the modulation index is increased, the amplitudes of side impulses increase proportionally, as do the existence of additional harmonics in the spectrum.

❖ Soft Keying

Using the same connections as in previous sections, this part of the experiment involved configuring the function generator to produce a square wave with a frequency of 1000 Hz, an amplitude of 5V (V_{ss} of 10V), and a duty cycle of 50%. The modulator mode was set to Digital (DIG), and Amplitude Shift Keying (ASK) with soft keying and a modulation index of 100% (A_12) was selected. The Cassy sensor UA1 was connected to the unipolar square-wave signal at socket MODIN, while the Cassy sensor UB1 was connected to the ASK signal at the output TXOUT of the modulator.

The measurement process was initiated, and was showed as follow:

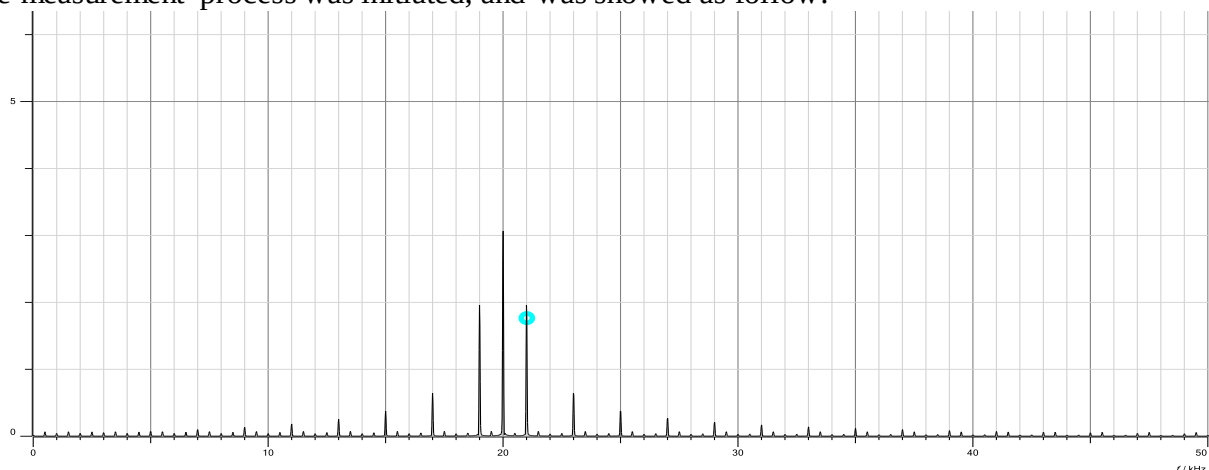


Figure 15: ASK message signal in frequency domain $m = 100\%$ soft keying

Soft keying has a smaller bandwidth when compared to hard keying. While the carrier frequency stays constant, the amplitude of the spectrum impulses in soft keying varies according to the amplitude fluctuations in the message signal. The impulses in the spectrum still reflect the harmonic components of the modulated signal, but their amplitudes fluctuate continually owing to the soft keying approach.

In a repeated iteration of the previous part, the experiment employed Amplitude Shift Keying (ASK) with soft keying and a modulation index of 50% (A_22). The measurement process was initiated to observe and analyze the outcomes as follow:

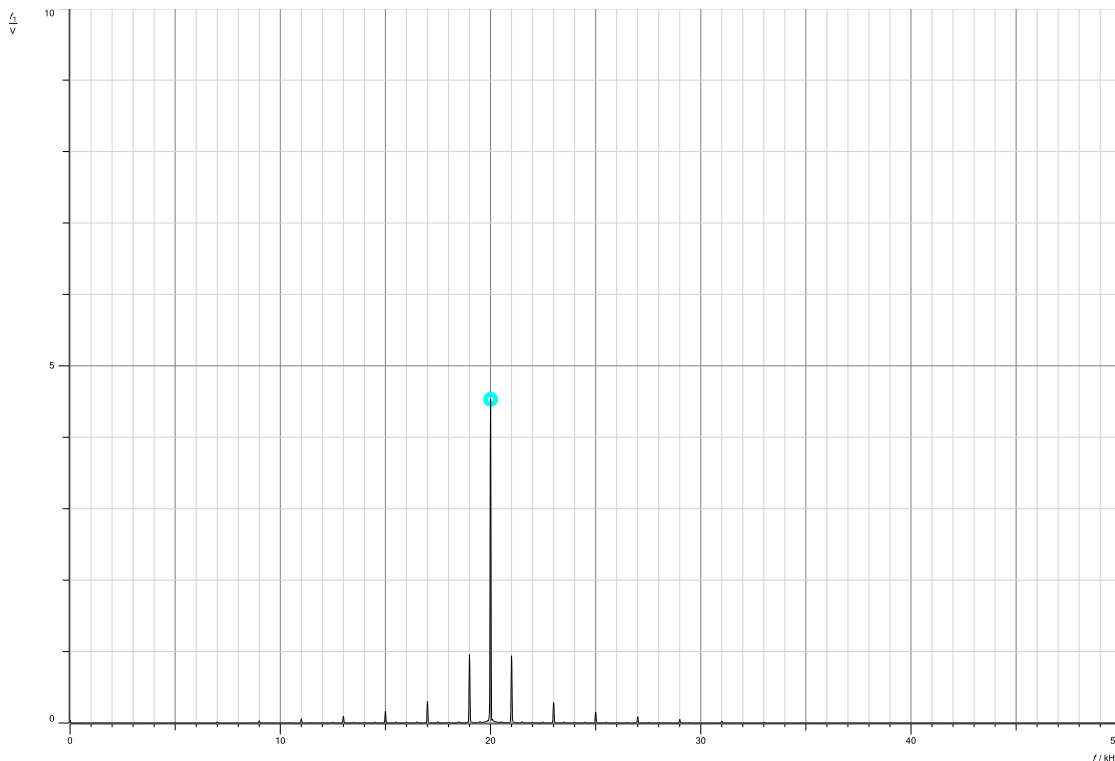


Figure 16: ASK message signal in frequency domain $m = 50\%$ soft keying

In soft keying, lowering the modulation index to 50% increases the amplitude of the carrier impulse while decreasing the amplitudes of the spectrum impulses. This decrease in modulation index leads to a smaller spectrum bandwidth.

In general, the spectrum bandwidth of Amplitude Shift Keying (ASK) modulation in hard keying is larger than that of soft keying. The occurrence of high-frequency harmonics created by quick transitions in hard keying techniques is attributed to this.

❖ Studying the effect of changing the message signal amplitude on the spectrum

Continuing with the same setup and connection as in previous sections, the experiment proceeded by reducing the message signal amplitude to 1.5V ($V_{ss} = 3V$). The ASK signal spectrum was obtained using Cassy Lab software, and a corresponding figure of the signal in the time & frequency domains were captured, as depicted below.

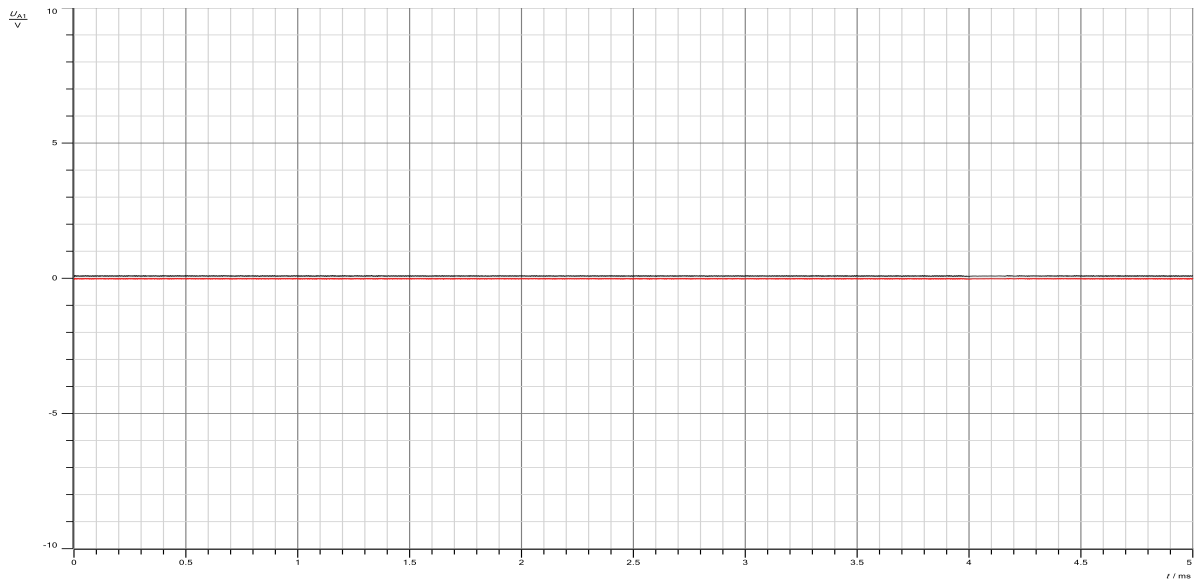


Figure 17: ASK signal Time Domain Reducing message amplitude

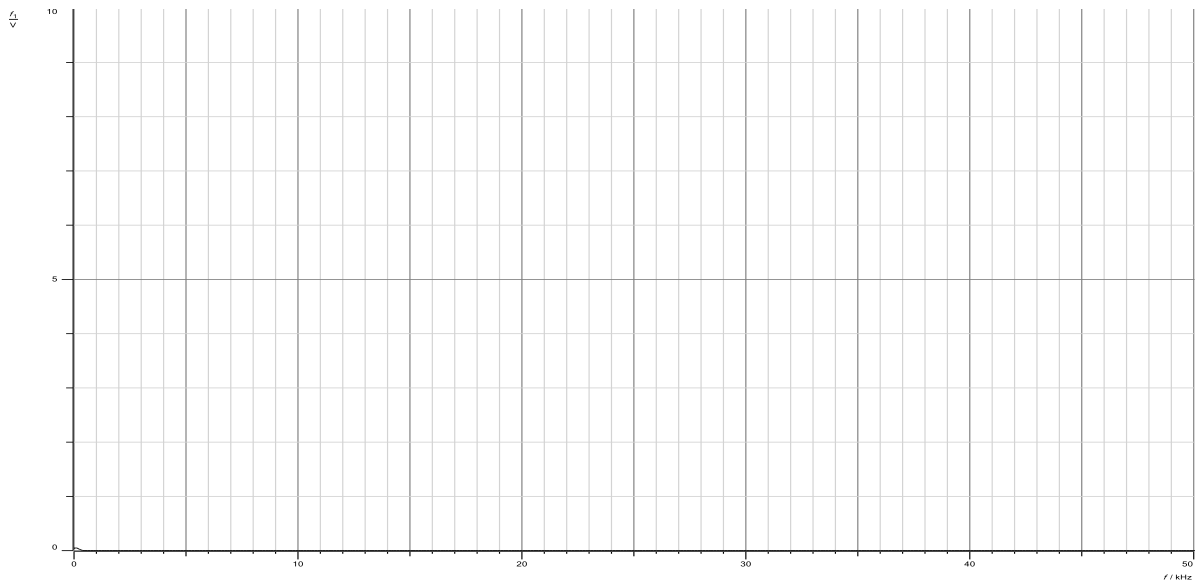


Figure 18: ASK signal Frequency Domain Reducing message amplitude

Due to the reduced amplitude of the signal at 1.5V, which falls below the lower threshold DC value of 1.68V, the signal was observed as zero. The lower amplitude resulted in the signal being perceived as having no significant presence or activity.

❖ Studying the effect of changing the message signal frequency on the spectrum

The experiment required setting the function generator to create a square wave with a frequency of 500 Hz, V_{ss} of 10V, and a duty cycle of 50%, continuing with the consistent setup and connection from previous parts. Cassy sensor UA1 was attached to the unipolar square-wave signal at MODIN, while Cassy sensor UB1 was connected to the ASK signal at the modulator's output TXOUT. The ASK signal spectrum was produced using Cassy Lab software, and a time & frequency domains images of the signal was taken, as shown below.

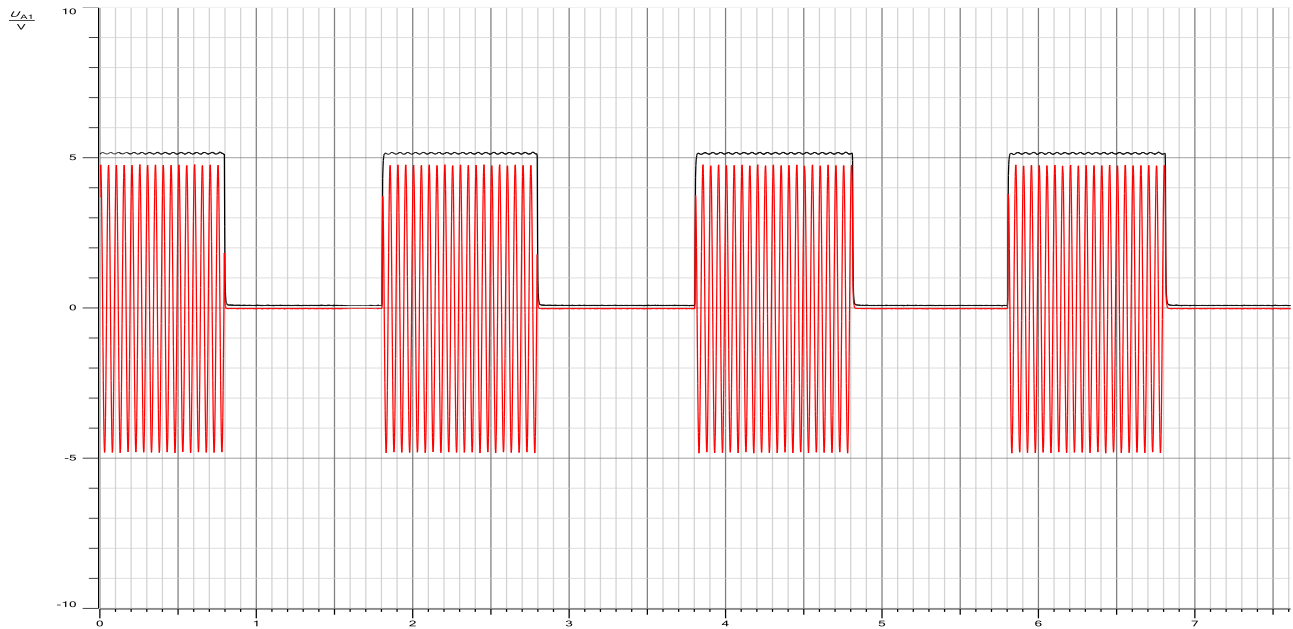


Figure 19: ASK modulated Signal time domain reducing message frequency

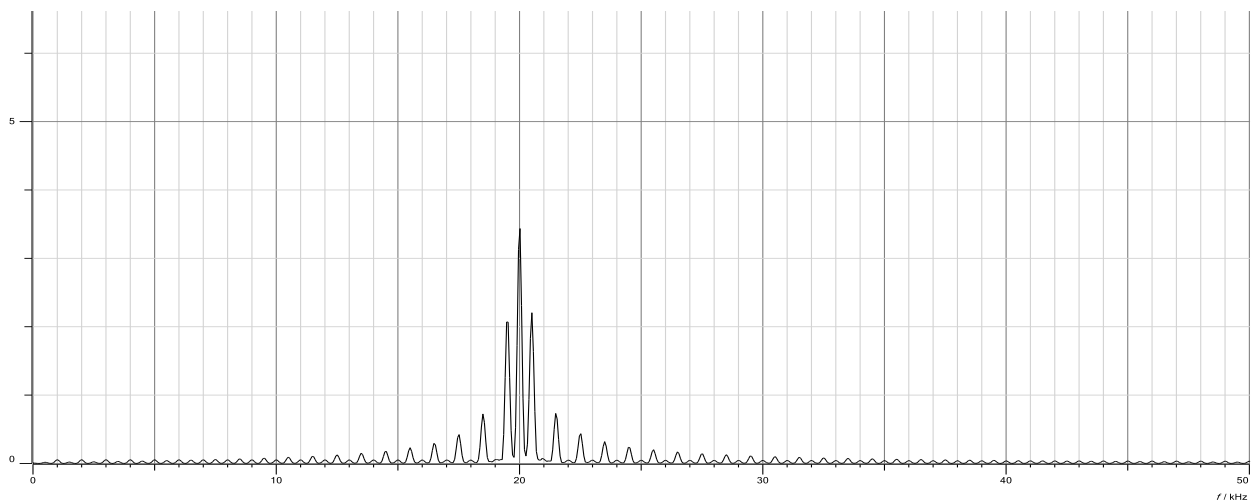


Figure 20: ASK modulated Signal frequency domain reducing message frequency

Reducing the frequency of the message signal in ASK modulation leads to a narrower bandwidth and longer period of the modulated signal.

❖ Studying the effect of changing the message signal duty cycle on the spectrum

Continuing with the consistent connection from the previous part, the experiment involved configuring the function generator to generate a square wave with a frequency of 1KHz, V_{ss} of 10V, and initially, a duty cycle of 10%. The duty cycle was then increased, and the impact on the Amplitude Shift Keying (ASK) spectrum was observed. The ASK signal spectrum was plotted using Cassy Lab software, and a corresponding figure of the signal in the time & frequency domain was captured, as depicted in below.

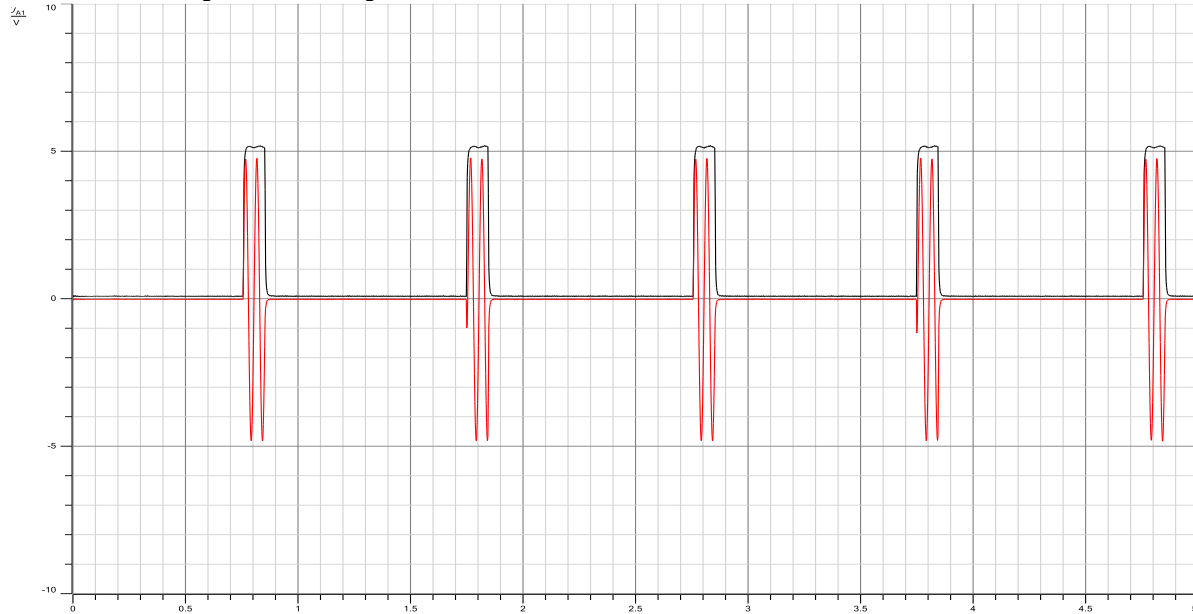


Figure 21: ASK modulated Signal time domain reducing duty cycle

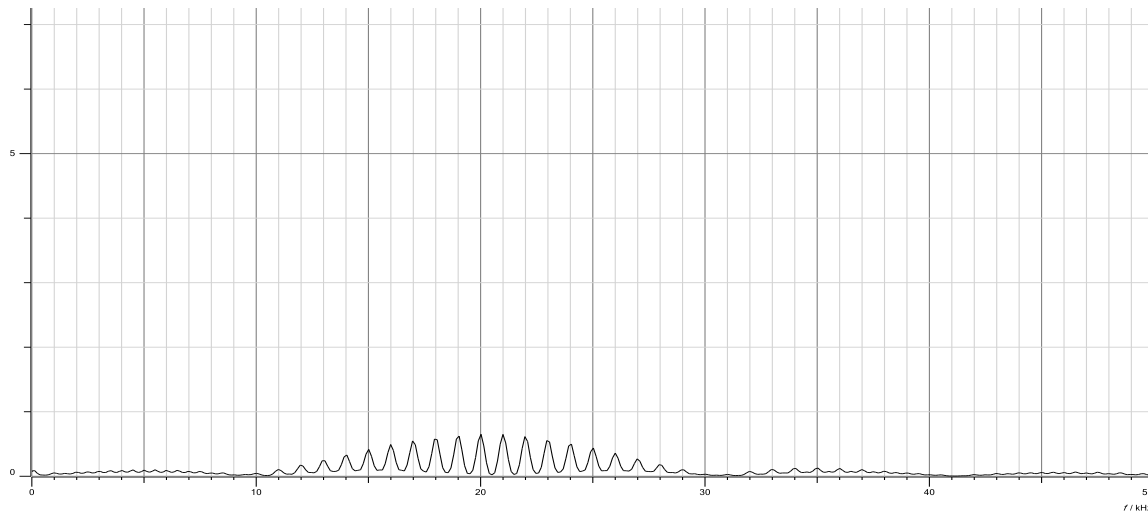


Figure 22: ASK modulated Signal frequency domain reducing duty cycle

Varying the duty cycle of the message signal in ASK modulation affects the active and inactive states' duration in the time domain. Reducing the duty cycle to 10% shortens the active state duration and weakens the amplitudes of odd harmonics in the spectrum. The ASK signal spectrum shows changes in harmonic amplitudes and the presence of even harmonics while maintaining a consistent fundamental frequency and harmonic spacing.

Part 3: Amplitude shift keying Demodulation (Time domain)

The components were assembled according to the following configuration:

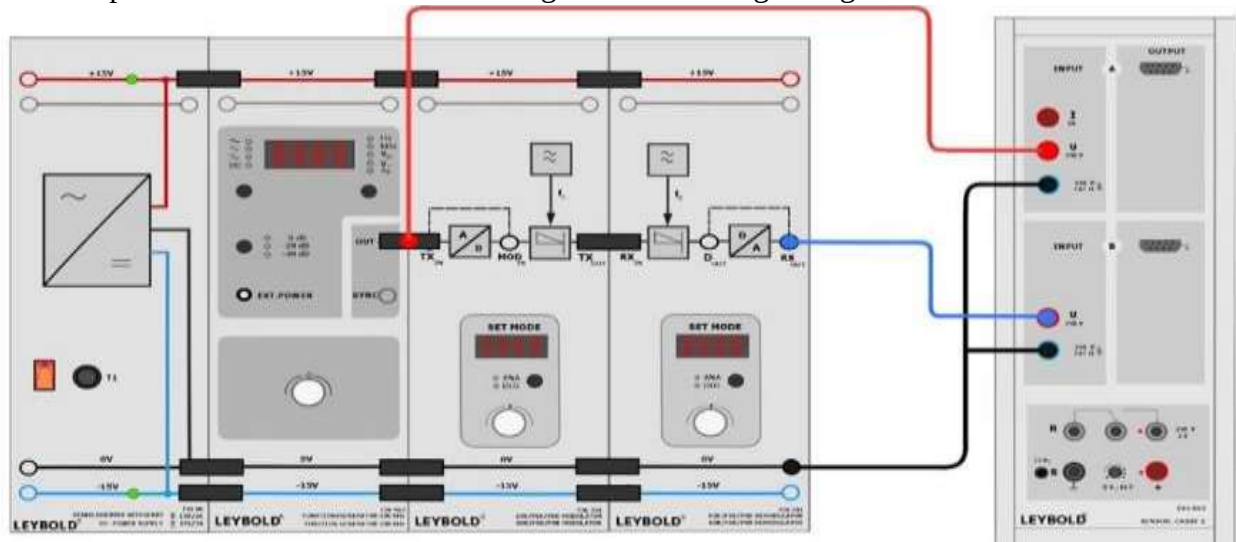


Figure 23: Assembling Components for ASK Modulation

❖ Hard Keying:

The measurements were initiated with the modulator mode set to Digital (DIG) and Amplitude Shift Keying (ASK) using hard keying and a modulation index of 100%(A₁₁). The resulting output is displayed below.

➤ At time domain:

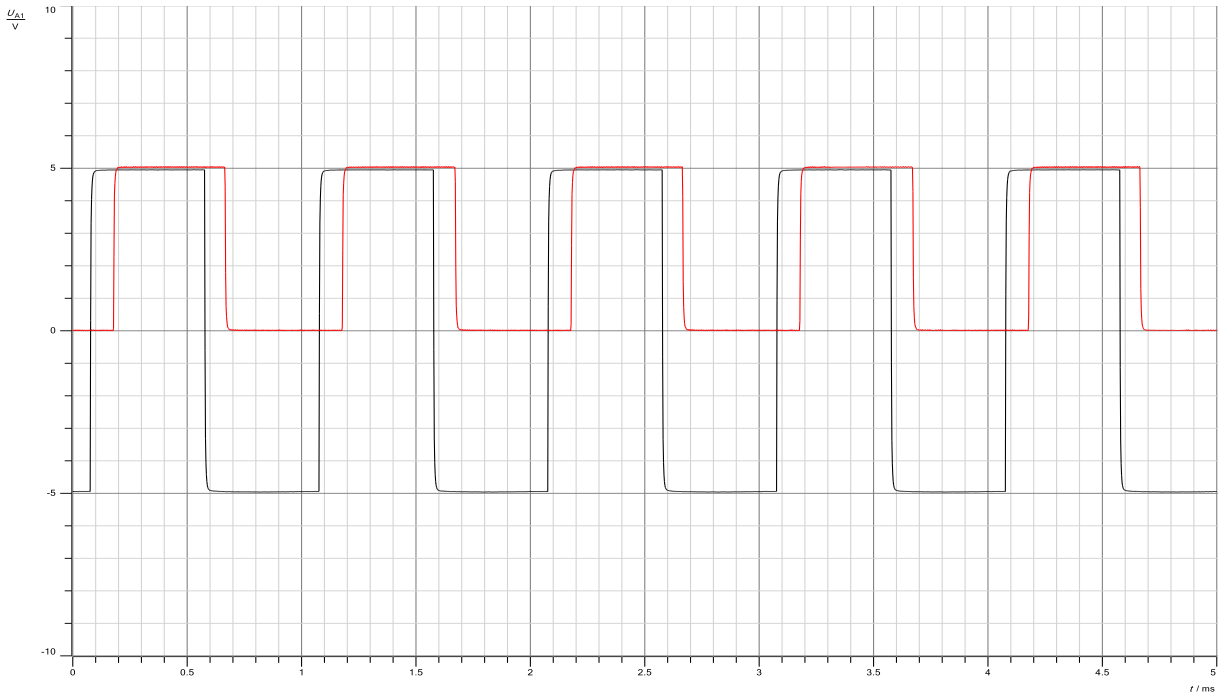


Figure 24: ASK demodulated signal time domain hard keying $m = 100\%$

➤ At frequency domain:

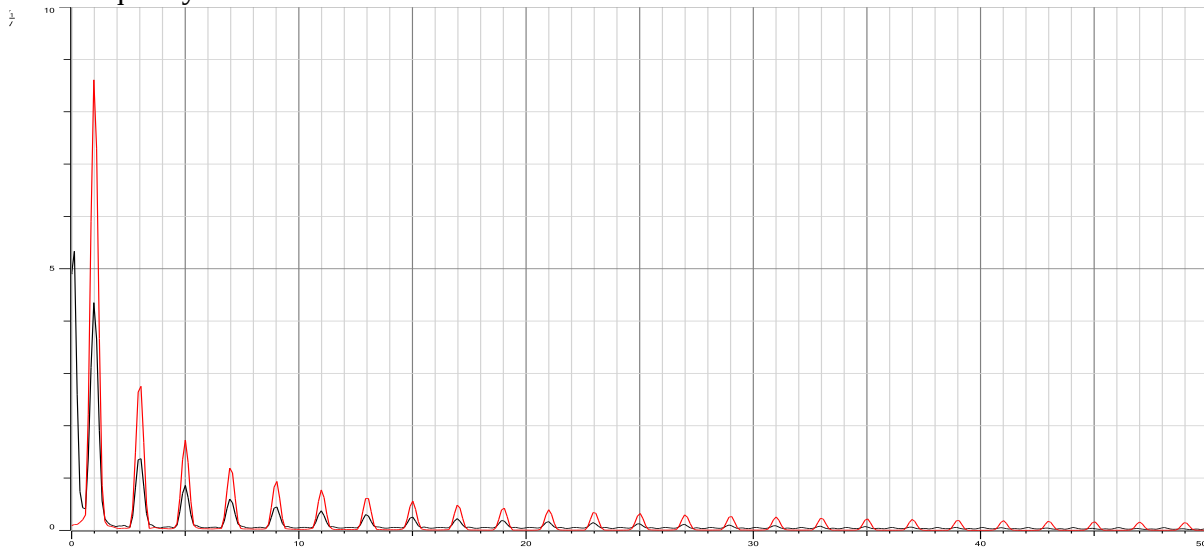


Figure 25: ASK demodulated signal frequency domain hard keying $m = 100\%$

When the results with a modulation index of 100% are examined, it is clear that the demodulation procedure is effective. In both the temporal and frequency domains, the demodulated signal substantially matches the original message signal. This shows that the ASK modulation and demodulation techniques employed in this experiment are effective in recovering the message signal precisely.

The measurements were started after selecting the Amplitude Shift Keying (ASK) mode with hard keying and a modulation index of 50% (A_21). The outputs, shown in Figures [26&27], shows the modulation process in operation. The ASK signal has noticeable amplitude changes.

➤ At time domain:

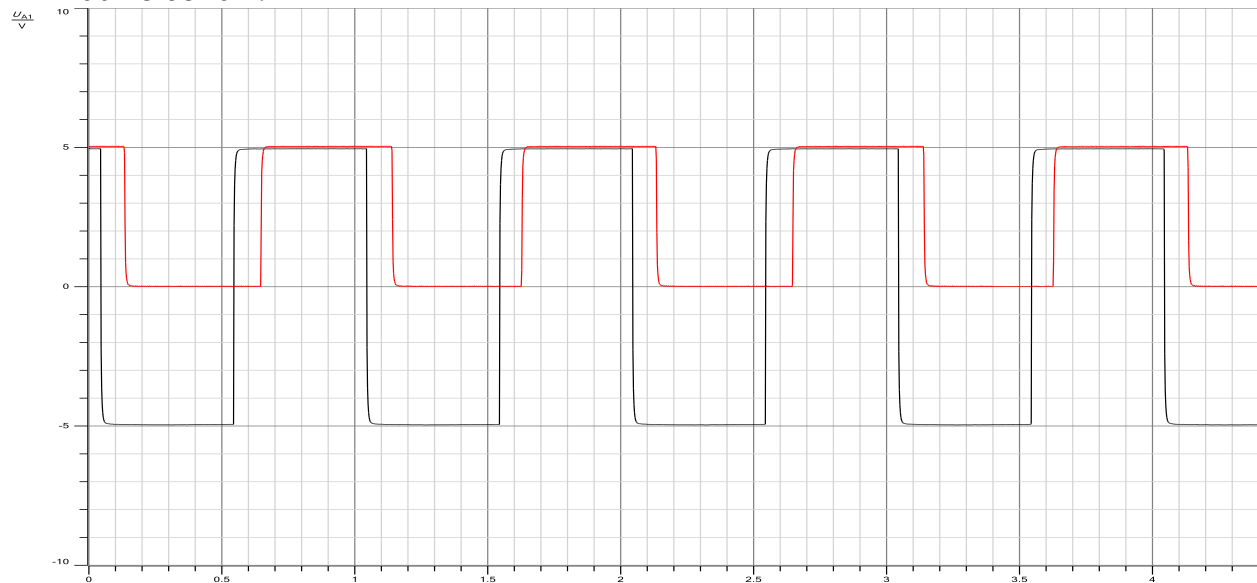


Figure 26: ASK demodulated signal time domain hard keying $m = 50\%$

➤ At frequency domain:

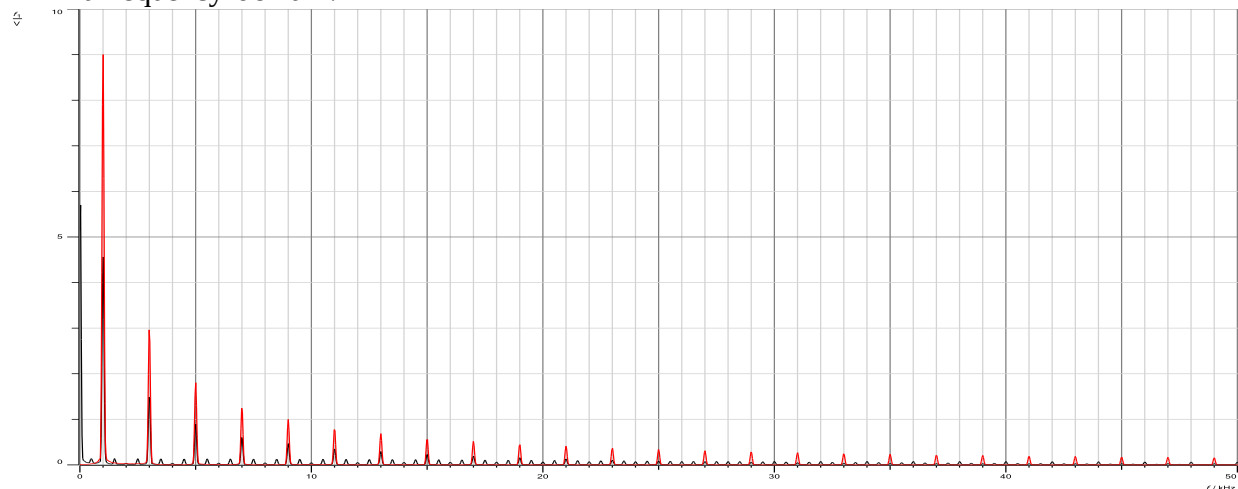


Figure 27: ASK demodulated signal frequency domain hard keying $m = 50\%$

Results with a modulation index of 50% show that the demodulation method is successful. The demodulated signal is time and frequency aligned with the original message signal. That ASK with a modulation index of 50% may have a poorer signal-to-noise ratio than ASK with 100% hard keying. The amplitudes of the carrier and message impulses in the spectrum show this disparity.

❖ Soft Keying

Keeping with the previously established connection arrangement, the experiment selected the Amplitude Shift Keying (ASK) mode with soft keying and a modulation index of 100% (A_12). The measuring procedure was initiated, and the resulting output is depicted below.

➤ At time domain:

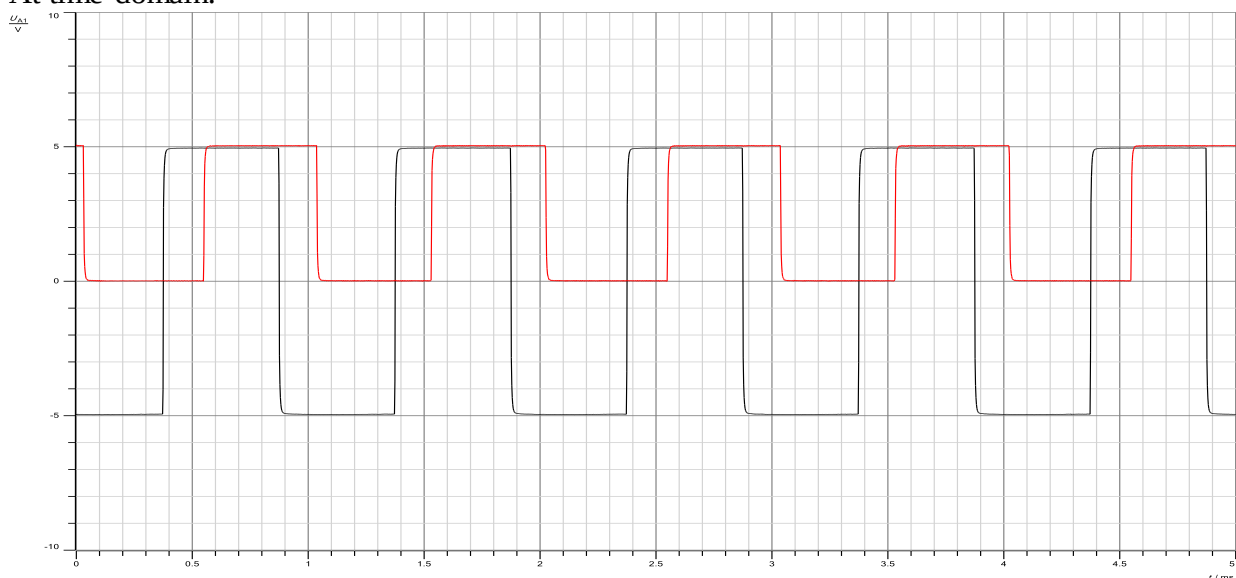


Figure 28: ASK demodulated signal time domain soft keying $m = 100\%$

➤ At frequency domain:

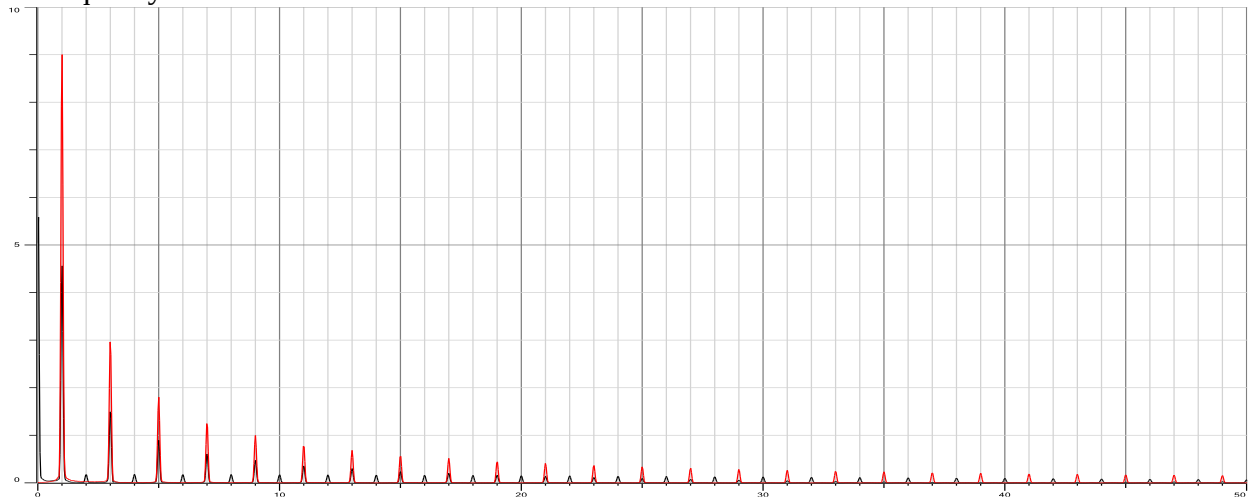


Figure 29: ASK demodulated signal frequency domain soft keying $m = 100\%$

The demodulation operation was clearly effective based on the obtained findings for the modulation index of 100%. In both the temporal and frequency domains, the demodulated signal is exactly aligned with the original message signal. This implies that the modulating information was correctly retrieved during the demodulation operation.

The measurements were started after selecting the Amplitude Shift Keying (ASK) mode with soft keying and a modulation index of 50%(A_22) and the output appeared as below.

➤ At time domain:

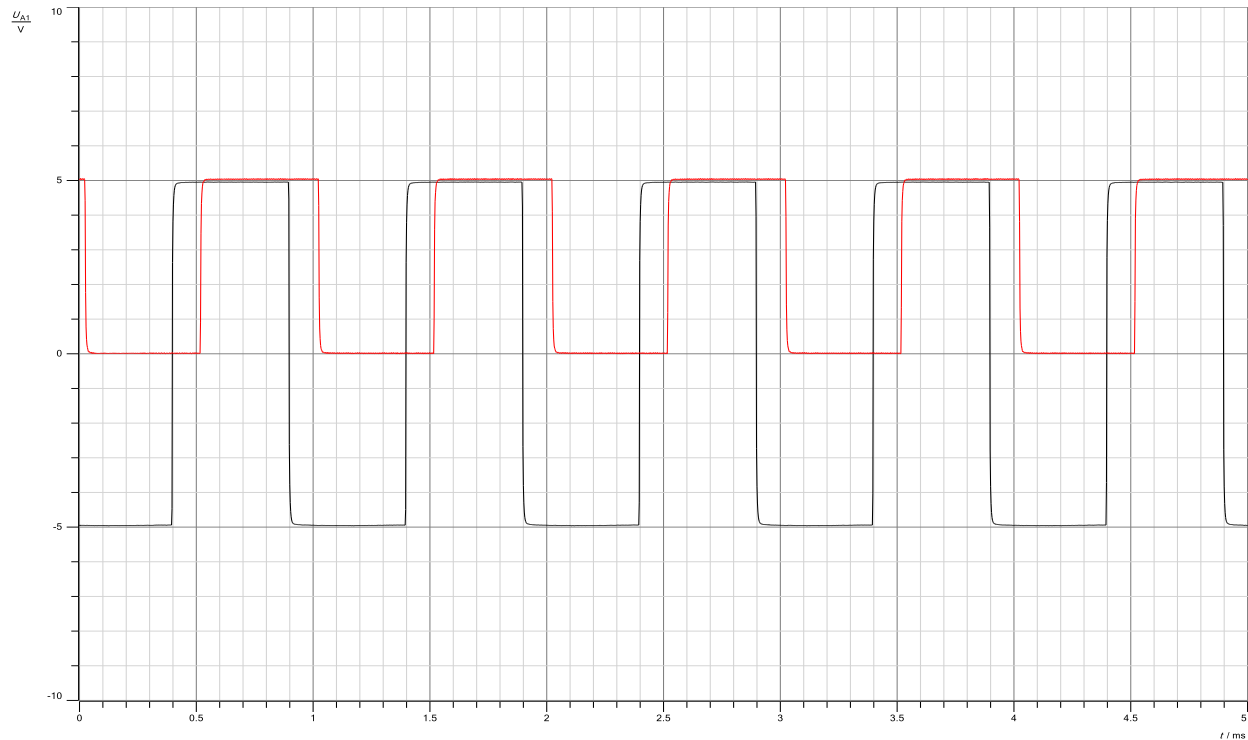


Figure 30: ASK demodulated signal time domain soft keying $m = 50\%$

➤ At frequency domain:

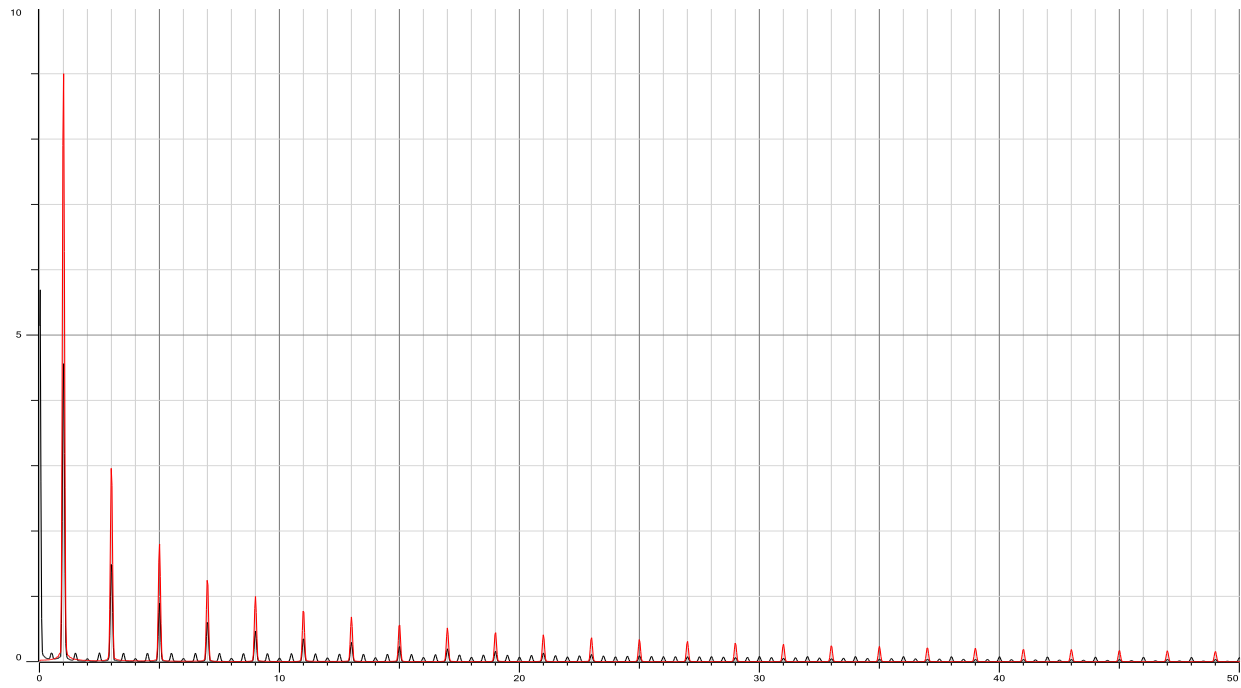


Figure 31: ASK demodulated signal frequency domain soft keying $m = 50\%$

When the results for $m = 50\%$ are examined, it is clear that the demodulation procedure was effective since the demodulated signal closely matches the original message signal in both the time and frequency domains. However, it should be noted that ASK with a modulation index of 50% may have a poorer signal-to-noise ratio than ASK with 100% soft keying, particularly in terms of carrier to message impulse amplitudes.

The demodulated signal attempts to match the original message signal, but noise and interference might create variations. The signal-to-noise ratio (SQNR) is the major difference between the message and the demodulated signals. The demodulated signal contains noise, which decreases its quality. As a result, optimizing the SQNR is critical for reducing the influence of noise on demodulation.

In addition, due to processing time, propagation delay, and extra demodulation stages, there is a temporal delay between the message signal and the demodulated signal. Because of its smoother modulation and higher processing needs, soft keying often has a greater time delay than hard keying. Notably, the modulation index has no direct effect on the time delay, however it may cause a slight, unnoticeable delay.

❖ Studying the effect of changing the message signal frequency on the demodulation

The previous portions' connection was maintained. However, the frequency of the message signal was lowered to 500 Hz, and a time domain image of the signal was captured, as shown below.

➤ At time domain:

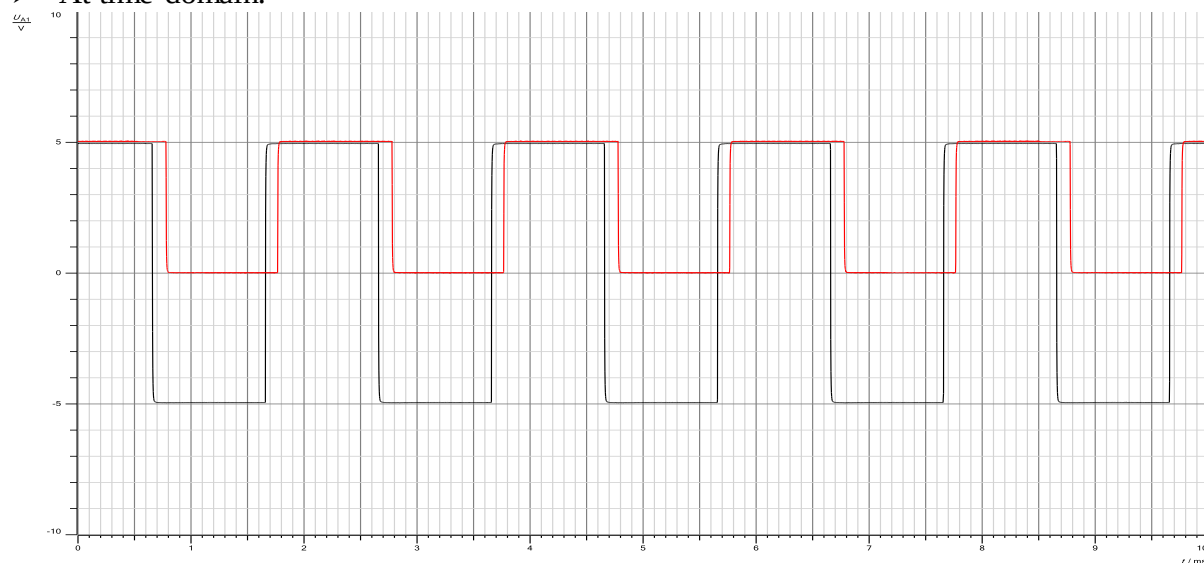


Figure 32: ASK demodulated Signal time domain reducing message frequency

➤ At frequency domain:

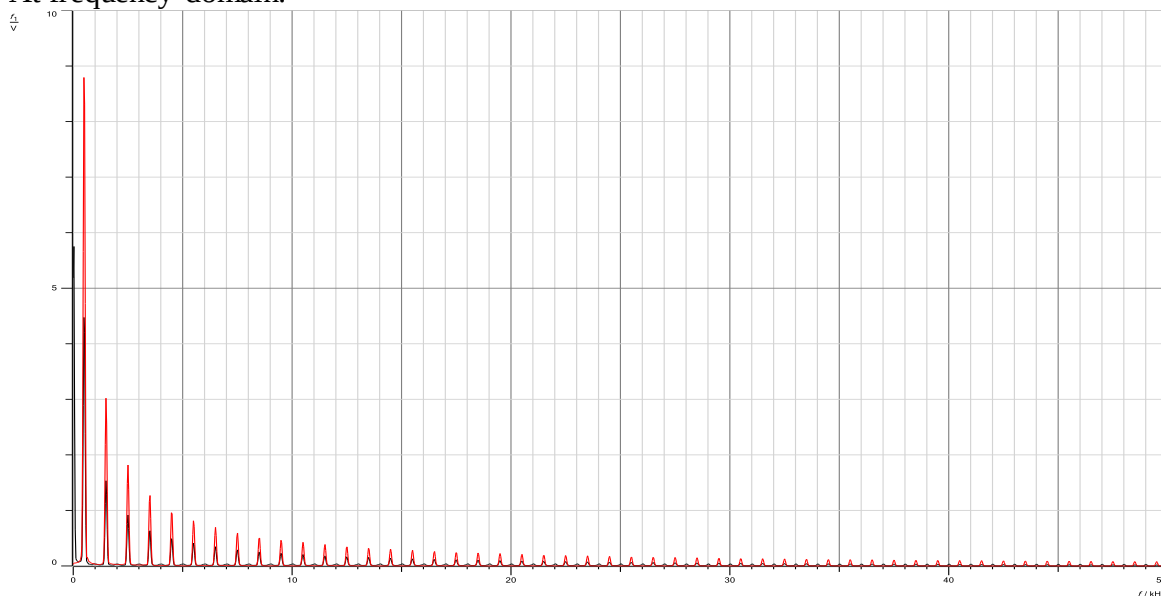


Figure 33: ASK demodulated Signal frequency domain reducing message frequency

Reducing the frequency of the message signal narrows the ASK modulated signal bandwidth and increases its period. Increasing the frequency has the reverse effect of increasing the bandwidth while decreasing the period.

❖ Studying the effect of changing the message signal duty cycle on the demodulation

The previous portions' connection was maintained. However, the frequency of the message signal was reset to 1KHz, the duty cycle was adjusted to 10%, and a time & frequency domains image of the signal was captured, as shown below.

➤ At time domain:

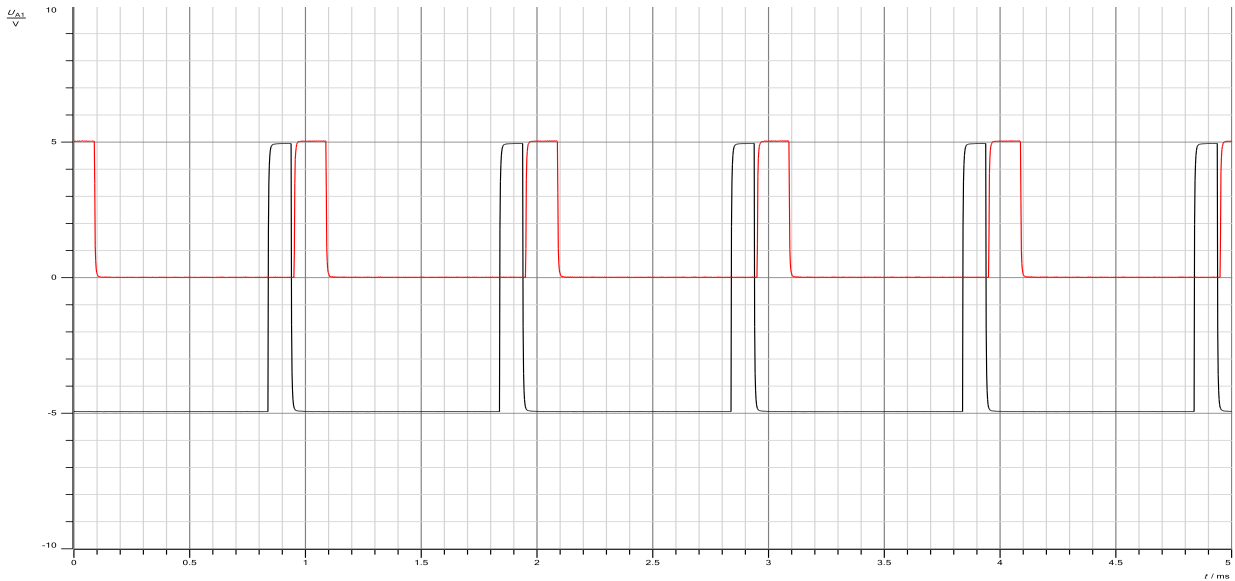


Figure 34: ASK demodulated Signal time domain reducing message duty cycle

➤ At frequency domain:

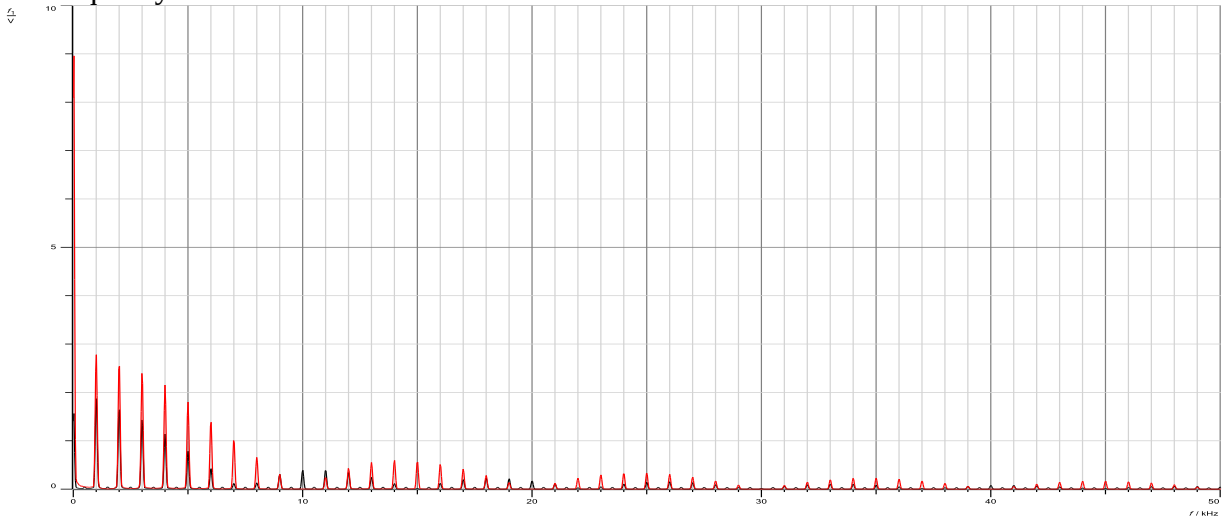


Figure 35: ASK demodulated Signal frequency domain reducing message duty cycle

Changing the duty cycle of the ASK waveform affects the active and inactive states in the time domain. A duty cycle of 50% is commonly preferred for ASK modulation as it balances amplitude variations and optimizes power spectrum utilization. Smaller duty cycles shorten the active state, while larger duty cycles lengthen it, impacting amplitude variations. These variations can impact signal transmission and demodulation processes.

4. Conclusion

In this experiment, the goal was to gain a deeper understanding of Amplitude Shift Keying (ASK) modulation, specifically through the investigation and comparison of Hard Keying and Soft Keying methods. Factors such as the modulation index, duty cycle, frequency, and amplitude of the message signal were explored to analyze the behavior of ASK signals in both the time and frequency domains. The experiment aimed to enhance our comprehension of ASK modulation and its intricacies, contributing to our knowledge of passive voice communication. By the end of the experiment, the objective of acquiring information about ASK and its practical implications was successfully achieved.

5.References

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